Appendix A

Specification

Husky Computers Ltd reserve the right to alter the specification or conditions of any product or service without prior notice.

A.1 Physical

Construction	Die-cast aluminium alloy.
Size	216.5mm x 157mm x 35.5mm.
Weight	(Including batteries) 1200 grams.
Sealing	Waterproof against accidental immersion.
Straps	Wrist strap on either side.

A.2 Fascia

Screen	240 x 64 pixel full graphics liquid crystal display with keyboard controlled contrast adjustment. Display area: 175mm x 50mm.
Keyboard	68 keys in standard QWERTY layout, with numeric keypad. All keys reprogrammable. Fully waterproof sealed rubber keys.
Backlight	Electroluminscent panel, with keyboard and program control.

A.3 Programming

Operating system	MS-DOS 3.3, ROM based.
Operating system interface	Assembly language access to standard and extended functions, via BIOS/MS-DOS.
Programming language	GWBASIC interpreter, ROM based.
Storage	Operating system and GWBASIC in ROM. Programs and data stored in 'permanent' RAM disk (battery backed), or optionally in user EPROM or Oracle GT disk drive.
Screen handling	ANSI/IBM PC interface, emulating IBM CGA text and graphics modes. The LCD screen forms a window on a larger (CGA-compatible) virtual screen. The window moves automatically and can also be moved by using the keyboard or under program control.
Keyboard	All keys reprogrammable in software.
Sound	IBM PC compatible, with mute function.

A.4 Communications

Ports	Two asynchronous RS232/V24 serial ports.
Port 1	COM1. Full modem interface, via standard 25 pin D type connector.
Port 2	COM2. Simple (3 wire plus CTS) interface, via a 5 pin FISCHER connector.
Configuration	Both ports emulate IBM PC ports, but additional facilities may be selected under software control, including:
Baud rate	Both ports, 110 to 38.400 Baud asynchronous.
Buffering	Independent 256 byte buffers for Tx and Rx for each port, interrupt-driven for efficiency.
Protocols	XON/XOFF protocol provided as standard for flow control.
Handshaking	RTS, CTS, DTR, DSR, RI, DCD, all selectable by software.
Options	Full synchronous interface to support SDLC and Bisynchronous communications. Optional protocols, including HWK3780 and MNP.

A.5 Screen

Text modes	In text modes, 25 x 40 or 25 x 80 characters on the virtual screen, up to 10 x 60 visible at a time in the LCD window. Variable text size, to enhance readability.
Graphics resolution	320 x 200 or 640 x 200 on the virtual screen. 240 x 64 pixels visible at a time in the LCD window.
Graphics text	In graphics modes, six different character sizes provide up to 10 x 60 on the LCD screen.
Graphics commands	Software support for drawing lines, boxes, ellipses, circles and points.
LCD facilities	Automatic scrolling over virtual screen, with keyboard and program control. Split screen facility. Screen lookup facility.

A.6 Memory

Types	Low power CMOS RAM and EPROM.
Retention	Battery support with fixed internal backup.
RAM	Minimum 1 MByte.
Standard EPROM	128 KByte, including: MS-DOS operating system Utilities GWBASIC interpreter
User EPROM	Various sizes available of EPROM and Flash EPROM, for user files.

A.7 Clock/calendar

Type	Software accessible time, day, date.
Accuracy	200ppm worst case, typically 2 seconds per day.
Integrity	Non-volatile, battery backed.

A.8 Microprocessor

Type	16 bit 8088-compatible.
Clock speed	8 MHz.

A.9 Batteries

Main batteries	Four 'AA' cells, Alkaline Manganese or Nickel Cadmium rechargeable.
Operating life	Typical operating lifetime of Alkaline Manganese batteries is up to 35 hours. Typical operating lifetime of Nickel Cadmium batteries (ie time between recharges) is somewhat less.
Power monitoring	Optional Advanced Power Mode maintains percentage charge remaining, accessible by user or application software. Automatic power conservation and shut off facilities.
Recharging	NiCads can be recharged internally via the supplied AC adaptor. Charging is under software control. (A fast charger is also available.)
Data retention	Built-in trickle-charged battery guarantees data integrity for at least one month when all other power removed (under normal operating conditions).